

Sanjana S S

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SUMMARY

Software Engineer(GenAI) with 2+ years of product engineering experience building **Python** backend services, ML pipelines, and **Generative AI applications** for large-scale, client-facing analytics platforms. Delivered end-to-end **Gen AI over financial documents** hands-on with **LLM orchestration (LangGraph, LangChain)**, **RAG over vector databases (ChromaDB, Snowflake DCR)**, **Model Context Protocol (MCP)**, and **React.js + FastAPI** AI dashboards with **RAGAS-based LLM evaluation**. Skilled in **containerized microservices, GitHub Actions CI/CD, and MLOps for LLMs**, with focus on **responsible AI, security, and cost-optimized inference**. Certified **Snowflake SnowPro Core** professional.

EXPERIENCE

ADA Global

Data Scientist / Software Engineer(GenAI)

Bengaluru, India

June 2024 – Present

FinDataAccelerator – Generative AI Data Pipeline Platform

2025 – present

Python, LangGraph, LangChain, ChromaDB, MCP, FastAPI, React.js, RAGAS, Docker, GitHub Actions, Groq, Ollama, Gemini

- Architected a **4-agent LangGraph multi-agent pipeline** (Ingestion → Quality → Transform → RAG) that processes **structured CSV/Parquet and unstructured PDF/text** financial documents from S3, replacing ad-hoc ETL scripts with a production-grade Gen AI data acceleration platform.
- Implemented a **RAG (Retrieval-Augmented Generation) agent** using **ChromaDB vector embeddings and semantic search** to enable natural-language querying over financial documents, achieving **sub-2s retrieval on 50K+ document chunks** and demonstrating efficient document retrieval at scale.
- Integrated a **Model Context Protocol (MCP) server layer** with **S3, PostgreSQL, and Filesystem connectors**, enabling LLM agents to securely access enterprise data sources without hardcoded credentials aligned with **responsible AI and security best practices for AI applications**.
- Built a **React.js + FastAPI** dashboard with real-time pipeline monitoring, a **RAG chat interface**, and an **LLM evaluation panel** surfacing **RAGAS metrics (faithfulness, answer relevancy, context precision)** applying **UI/UX patterns** designed specifically for AI interfaces.
- Designed a **cost-optimized LLM inference layer** using **Groq (fast cloud inference)**, **Ollama/phi3 (local)**, and **Gemini (embeddings + fallback)**, and containerized all services with **Docker + docker-compose**, orchestrated via **Prefect** and shipped through a **GitHub Actions CI/CD pipeline** embodying **MLOps best practices for LLM applications**.

Data Clean Room Platform – Privacy-Preserving Multi-Party Analytics

2024 – 2025

Python, FastAPI, LangChain, Snowflake Data Clean Rooms, GCP Vertex AI, Docker, Celery, Redis, PostgreSQL, GitHub Actions, CI/CD

- Architected a **microservices-based** backend in **Python (FastAPI)** enabling secure, privacy-preserving multi-party data collaboration across data partners and clients in a **BFSI-grade** environment with **GDPR/CCPA compliance, K-anonymity ($k \geq 5$) validation**, and **SHA-256 PII hashing**.
- Designed a **RAG-based semantic search** pipeline for audience segmentation, leveraging **vector embeddings** and **Snowflake Data Clean Rooms** to perform cross-schema analytics across both **structured and unstructured datasets**.
- Built an **LLM-orchestrated enrichment engine** using **LangChain** for dynamic **JinjaSQL template generation**, applying **prompt engineering** for **natural-language-to-SQL conversion** with **SHA-256 template deduplication** and auto-approved caching.
- Engineered a multi-cloud activation pipeline integrating **Google Ads, Meta, Amazon DSP, and Trade Desk APIs** with **GCP Vertex AI**-powered match scoring and intelligent retry logic; orchestrated **multi-cloud storage (AWS S3, Google Cloud Storage, Azure Blob)** with **Snowflake external tables** across **GCP**.

- Containerized services with **Docker + Docker Compose**, **Celery** workers, **Redis** caching, and **PostgreSQL (AsyncPG)** with Alembic migrations; enforced **RBAC + JWT authentication**, audit middleware, and **Sentry** observability; shipped via **GitHub Actions CI/CD** with comprehensive **pytest** suites in **Agile scrums with globally distributed teams**, applying **MLOps and responsible AI practices** for LLM performance monitoring and cost-optimized query execution.

City Union Bank

Relationship Manager – Operations & Sales

Coimbatore, India

March 2023 – September 2023

- Owned customer relationships, sales targets, and day-to-day branch operations as a **Relationship Manager**, building hands-on retail banking and finance domain knowledge across deposit and lending products, cross-sell workflows, and customer servicing in a live **BFSI** environment.
- Gained practical exposure to KYC, AML, and RBI regulatory norm, risk-sensitive transaction handling, and branch-level controls, providing first-hand finance & BFSI domain expertise.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

Generative AI & LLM Orchestration: LangChain, LangGraph, CrewAI, OpenAI Agents SDK, AutoGen, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Prompt Engineering (Chain-of-Thought, Few-Shot, Structured Outputs), OpenAI APIs, Groq, Ollama, Google Gemini

Model Evaluation & MLOps for LLMs: RAGAS (Faithfulness, Answer Relevancy, Context Precision), LLM Performance Metrics, Prefect Orchestration, Model Deployment, Cost & Performance Optimization, Responsible AI, AI Application Security

Vector Databases & Semantic Search: ChromaDB, Vector Embeddings, Semantic Search, Document Retrieval Systems

Frontend (AI Interfaces): React.js, Tailwind CSS, UI/UX Patterns for AI Applications

Backend & Microservices: FastAPI, Django, REST APIs, Microservices Architecture

Cloud, Containerization & CI/CD: Azure Cloud, AWS cloud, GCP Vertex AI, Docker, docker-compose, GitHub Actions, CI/CD Pipelines, Git, GitHub

Databases & Data Engineering: PostgreSQL, Snowflake, DuckDB, ChromaDB, ETL Pipelines, Structured & Unstructured Data Processing, Data Cleaning, Data Transformation, Query Optimization, Pandas, NumPy

Machine Learning & Data Science: Regression, Classification, Clustering, Forecasting, Anomaly Detection, Feature Engineering, Scikit-learn, Matplotlib, Seaborn

EDUCATION

Great Lakes Institute of Management

Post Graduate Programme in Data Science and Engineering

Chennai, India

September 2023 – March 2024

Bharathiar University

MA English Literature

Coimbatore, India

September 2020 – June 2022

Avinashilingam Institute for Home Science & Higher Education

BA English Literature – Gold Medalist

Coimbatore, India

June 2017 – August 2020

CERTIFICATIONS

- **AI Engineer Agentic Track: The Complete Agent & MCP Course** – Udemy
- **AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents** – Udemy
- **SnowPro Core Certification** – Snowflake (Score: 95%)
- **SQL (Advanced)** – HackerRank

ACHIEVEMENTS

- **Gold Medal** – BA English Literature, Avinashilingam Institute
- **French DELF Level A1** – Alliance Française (85%); **German A1** – verbal and written communication